APPSC GROUP 1 / GROUP 2 — GEOGRAPHY

Volume 1: General Geography — explained from the basics

Earth and Solar System · Motions and Seasons · Time · Earth's Interior · Plate Tectonics · Rocks and Landforms · Atmosphere · Winds · Cyclones · Climate · Oceans

READ THIS FIRST

How to use this book (read once):

1. Read each chapter in order. Every topic has the same pattern: **What it is → How and why it happens → Diagram → Remember → AP / Kurnool link → Quick check.**
2. Do not try to memorise everything. First **understand the blue box (In simple words)**. Facts stick only after the idea is clear.
3. Colour code: **Blue** = simple explanation · **Orange** = Andhra Pradesh connection · **Red** = exam traps · **Green** = 2-minute revision · **Purple** = self-test.
4. Plan: one chapter per day (about 1.5-2 hours), then re-read only the green boxes after 3 days and again after 10 days.
5. In the Prelims, Geography is 30 of 120 questions in Paper I. Each wrong answer costs 1/3 mark. If you are not sure and cannot remove at least two options, leave it.

Note on facts: figures change (for example India's tiger count, Ramsar sites, coastline length). Where a number can change it is marked (verify). Recheck those in the Current Affairs notes close to the exam.

CHAPTER 1 — THE UNIVERSE AND THE SOLAR SYSTEM

Start here: where does the Earth sit?

1.1 What is what

IN SIMPLE WORDS (what / how / why)

Universe = everything. It has billions of **galaxies**. A **galaxy** = billions of stars held together by gravity. Ours is the **Milky Way** (spiral shape).

A **star** (like the Sun) makes its own light and heat. A **planet** goes around a star and shines only by reflecting its light. A **satellite** goes around a planet (Moon is Earth's satellite).

The **Solar System** = Sun + 8 planets + moons + dwarf planets + asteroids + comets.

Big Bang theory (accepted explanation for the origin of the universe): about 13.8 billion years ago everything started from one hot dense point and has been expanding since.

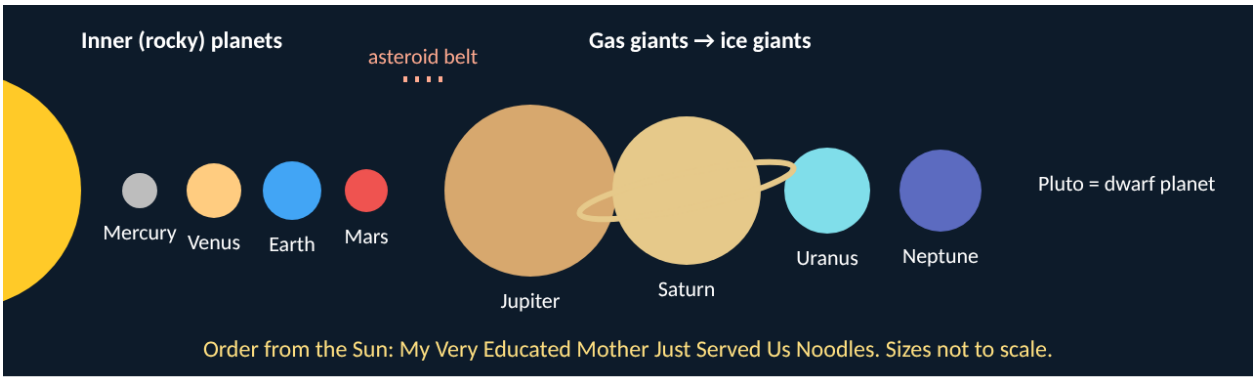


Fig 1.1 The planets in order (sizes not to scale)

1.2 Facts to remember

Planet	Key point for exams
Mercury	Closest to Sun; smallest planet; no moon; shortest year (88 days)
Venus	Hottest planet (thick CO_2 atmosphere); spins backwards (east to west); called Earth's twin; brightest 'star' (morning/evening star)
Earth	Only known planet with life; 'Blue Planet'; one moon; densest planet of the solar system
Mars	'Red Planet' (iron oxide); two moons (Phobos, Deimos); tallest volcano Olympus Mons
Jupiter	LARGEST planet; most massive; Great Red Spot (storm); many moons
Saturn	Famous rings; least dense (would float on water); second largest

Planet	Key point for exams
Uranus	Rotates on its side (tilted about 98°); coldest atmosphere
Neptune	Farthest planet; windiest

1.3 Why a planet and not a dwarf planet?

A **planet** (IAU 2006 definition): (1) orbits the Sun, (2) round because of its own gravity, (3) has **cleared its orbit** of other objects. Pluto fails (3), so it is a **dwarf planet** (others: Ceres, Eris, Haumea, Makemake).
Inner planets (Mercury, Venus, Earth, Mars) = small and rocky (terrestrial). **Outer planets** = big: Jupiter, Saturn are **gas giants**; Uranus, Neptune are **ice giants**. Between Mars and Jupiter lies the **asteroid belt**.

ANDHRA PRADESH / KURNOOL CONNECTION

ISRO, Sriharikota (Tirupati district, AP) is India's spaceport (Satish Dhawan Space Centre). PSLV, GSLV, SSLV, Chandrayaan launches happen here. Questions on space + AP often appear together.

EXAM TRAPS

Sun is a **star**, not a planet. Moon is a **satellite**, not a planet. Largest planet = Jupiter, but **densest** = Earth and **least dense** = Saturn.
Hottest planet is **Venus** (not Mercury, even though Mercury is closer).
Pluto = dwarf planet since 2006.

2-MINUTE REVISION

Order: Mercury, Venus, Earth, Mars | Jupiter, Saturn | Uranus, Neptune.
Venus hottest · Jupiter biggest · Mercury smallest and nearest · Neptune farthest · Mars red · Saturn rings.
Light from Sun reaches Earth in about 8 minutes 20 seconds; from Moon in about 1.3 seconds.

CHECK YOURSELF (answers at the end of the chapter)

Q1. Which planet rotates from east to west? **Venus** · Q2. Planet with the least density: **Saturn** · Q3. Pluto is a: **dwarf planet**

CHAPTER 2 — EARTH: SHAPE, MOTIONS AND SEASONS

Why do we have day and night, summer and winter?

2.1 Shape and size

Earth is not a perfect ball. It is slightly flattened at the poles and bulges at the equator, called a **geoid** (oblate spheroid).
Equatorial diameter about **12,756 km**; polar diameter about 12,714 km; equatorial circumference about **40,075 km**. About **71% water and 29% land**.
Earth is the **third planet** from the Sun; average distance about **149.6 million km** (= 1 Astronomical Unit).

2.2 Two motions

IN SIMPLE WORDS (what / how / why)

ROTATION = Earth spinning on its own axis (like a top). One turn = **24 hours** (23 h 56 min precisely). Direction: **west to east**. **Result: DAY and NIGHT.** (The Sun seems to rise in the east because we are moving eastward.)
REVOLUTION = Earth going around the Sun in an oval path (orbit). One round = **365¼ days**. That extra quarter day is why we add **1 day to February every 4 years (leap year)**. **Result: SEASONS and changing length of day.**
The axis is **tilted 23.5°** from the vertical (or 66.5° to the orbit plane) and always points towards the **Pole Star**. **The tilt is the real reason for seasons.**

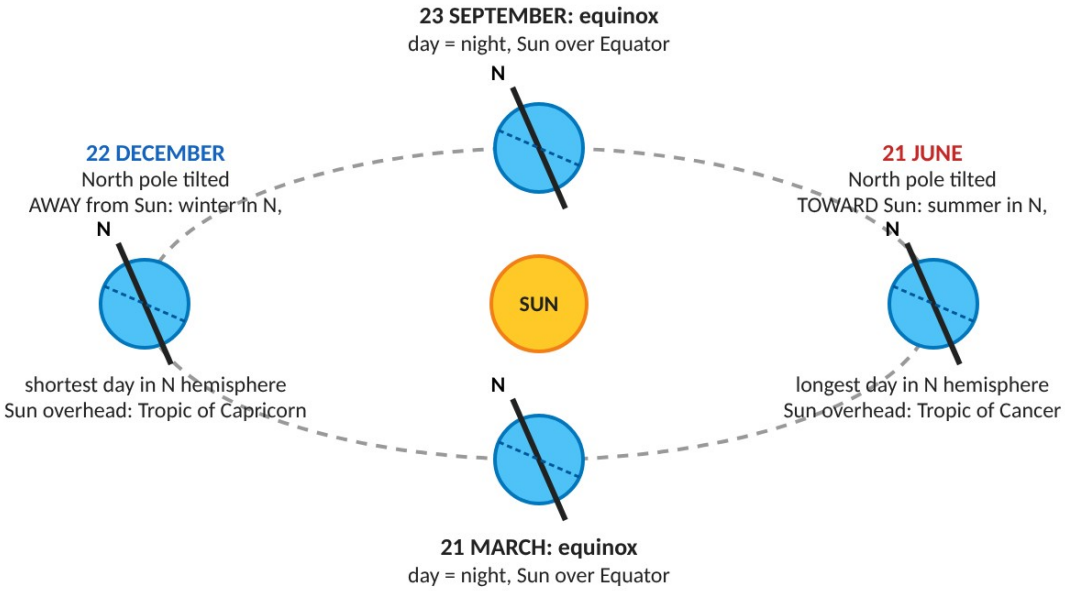


Fig 2.1 How the tilt makes seasons (top view of the orbit)

IN SIMPLE WORDS (what / how / why)

Think of it like this: in June the northern half leans **towards** the Sun. Sunlight falls almost straight down (concentrated, hot) and the days are long. Six months later the northern half leans **away**: sunlight comes slanting (spread out, weak) and days are short.
Common mistake: seasons are NOT caused by Earth being nearer to or farther from the Sun. In fact Earth is **nearest** to the Sun (**perihelion**) around **3 January** and **farthest** (**aphelion**) around **4 July**, the opposite of what people expect.

Date	Name	What happens
21 March	Spring (vernal) equinox	Sun directly over the **Equator**. Day = night everywhere
21 June	Summer solstice	Sun over **Tropic of Cancer (23.5°N)**. Longest day in the north. Arctic Circle has 24-hour daylight
23 September	Autumn equinox	Sun over Equator again. Day = night
22 December	Winter solstice	Sun over **Tropic of Capricorn (23.5°S)**. Shortest day in the north

2.3 Important lines on the globe

Equator 0° · Tropic of Cancer 23.5°N (passes through **8 Indian states**: Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram) · **Tropic of Capricorn 23.5°S · Arctic Circle 66.5°N · Antarctic Circle 66.5°S**.
Torrid zone (between the tropics, Sun overhead at noon at least once a year) · **Temperate zones** (tropics to circles) · **Frigid zones** (circles to poles).
Midnight Sun: near the poles the Sun stays above the horizon for weeks in summer.

ANDHRA PRADESH / KURNOOL CONNECTION

Andhra Pradesh is **south of the Tropic of Cancer**, entirely in the **torrid zone**, so it is hot nearly all year. The Sun is overhead around **mid-April to mid-May** (noon) in AP, which is why **April-May** are the hottest months here (before monsoon).

EXAM TRAPS

Rotation = day/night. Revolution = seasons. Do not swap.
Earth is nearest to Sun in **January**, not in summer.
Tropic of Cancer does NOT pass through AP, Karnataka, Maharashtra or Odisha.

2-MINUTE REVISION

Rotation: 24 h, west to east → day/night. Revolution: 365¼ days → seasons, leap year.
Tilt 23.5° → seasons. Equinox 21 Mar and 23 Sep. Solstice 21 Jun and 22 Dec.
Perihelion 3 Jan · Aphelion 4 Jul.

CHECK YOURSELF (answers at the end of the chapter)

Q1. On 21 June the Sun is overhead at: **Tropic of Cancer** · Q2. Cause of seasons: **tilt of axis + revolution** · Q3. Earth is nearest to Sun in: **January**

CHAPTER 3 — LATITUDE, LONGITUDE AND TIME

How does the clock change from place to place?

3.1 The grid

IN SIMPLE WORDS (what / how / why)

Latitudes are imaginary circles **parallel to the Equator** (0° to 90° N or S). They tell how far **north/south** a place is, and decide **climate**.
Longitudes (meridians) are half-circles joining the poles (0° to 180° E or W). **0° passes through Greenwich (London)** = **Prime Meridian**. They decide **time**.
Kurnool is at roughly **15.8°N, 78°E**. AP as a whole lies between about 12.6°N and 19.9°N and 76.8°E-84.8°E (undivided AP limits: verify).

3.2 Why time changes: the simple maths

IN SIMPLE WORDS (what / how / why)

Earth turns 360° in 24 hours → **15° every hour** → **1° = 4 minutes**.
Places **east** of Greenwich see the Sun earlier → their clocks are **ahead (add)**. Places **west** are **behind (subtract)**.
Local time = Sun-based time of a place. To avoid chaos, a country uses **one standard time**. **India: IST = 82.5°E meridian** (passes near Mirzapur, Uttar Pradesh), which is **5 h 30 min ahead of GMT**.
Worked example: When it is 12 noon in London (0°), time in India = 12:00 + 5:30 = **5:30 pm**. When it is 8 pm in India, London = 8:00 pm − 5:30 = **2:30 pm**.

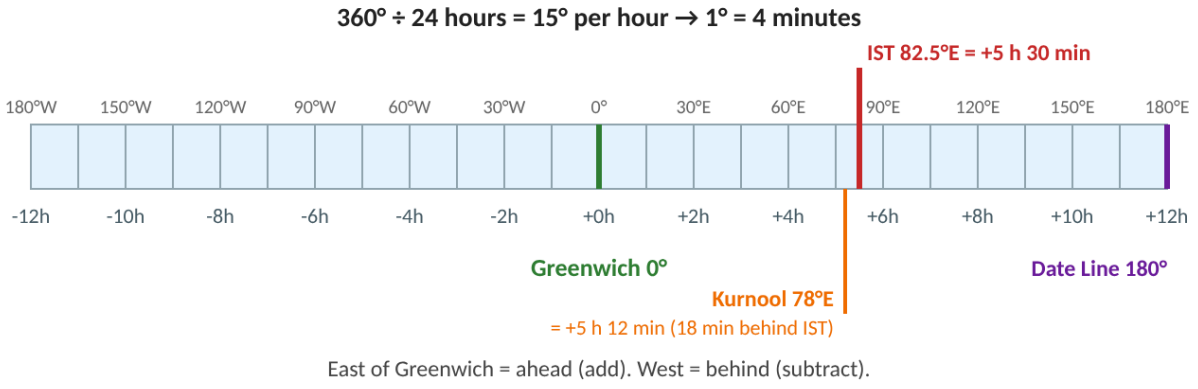


Fig 3.1 Time and longitude

International Date Line (IDL) follows roughly **180°**. Crossing it going **east** you go **back one calendar date**; going **west** you **skip forward one date**. It bends to avoid splitting countries.
India's width: about 30° of longitude from west (Gujarat) to east (Arunachal) → about **2 hours difference** in sunrise. This is why Assam tea gardens use 'garden time' and there is a debate on a second time zone.
Other big countries: Russia 11 time zones, USA (mainland) 4, France (with overseas territories) the most.

ANDHRA PRADESH / KURNOOL CONNECTION

Kurnool at 78°E is 4.5° west of the IST line → local time is about **18 minutes behind IST** (4.5 × 4 min). Sunrise in Kurnool is later than in Srikakulam (84°E) by about 24 minutes.

EXAM TRAPS

IST is based on **82°30′E**, not 80°E, and not the longitude of Delhi (77°E).
Crossing the Date Line **eastward** you **subtract** one calendar day (repeat a date); crossing **westward** you **add** one day (skip a date). Students often reverse this.

2-MINUTE REVISION

15° = 1 hour · 1° = 4 min · East add, West subtract · IST 82.5°E = GMT + 5:30.

CHECK YOURSELF (answers at the end of the chapter)

Q1. IST meridian: **82°30′E** · Q2. If it is 10 am GMT, IST is: **3:30 pm** · Q3. Time decided by: **longitude**; climate decided by: **latitude**

CHAPTER 4 — INTERIOR OF THE EARTH

What is under our feet?

IN SIMPLE WORDS (what / how / why)

We cannot dig to the centre (the deepest drill is about 12 km; radius is **6,371 km**). So scientists study **earthquake waves**. Waves

change speed and direction when they meet different materials, just like light bends in water.
Inside, **temperature and pressure rise with depth**. The Earth has 3 layers: crust (thin skin), mantle (thick middle, hot flowing rock), core (iron-nickel centre).

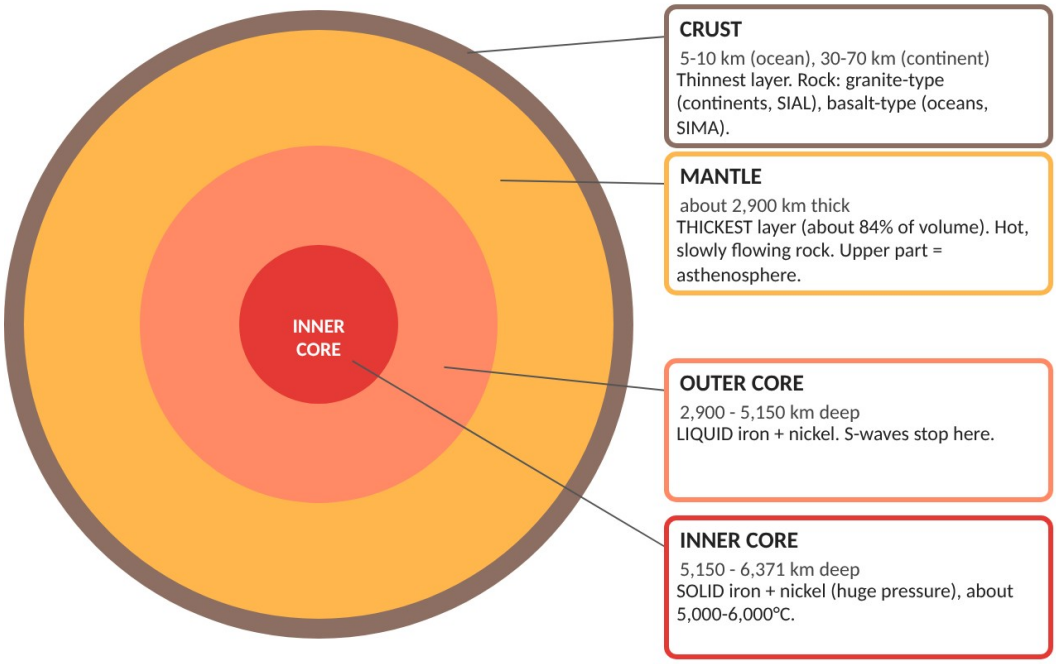


Fig 4.1 Layers of the Earth

Lithosphere = crust + the very top (rigid) of the mantle. It is broken into **plates**. Below is the soft **asthenosphere** on which the plates float and move.
Rocks of the crust: continents mainly **granite-type (SIAL = silica + aluminium)** — lighter; ocean floors **basalt-type (SIMA = silica + magnesium)** — heavier.
Core = NIFE (nickel + iron). Earth's **magnetic field** comes from the moving liquid outer core.

HOT SPOTS

Why do scientists say the outer core is liquid? **S-waves (secondary waves) cannot pass through liquids** and disappear there (shadow zone). **P-waves** pass through both solids and liquids but bend.
Order of arrival at a station: **P first (Primary), then S, then surface (L) waves**.

2-MINUTE REVISION

Crust thinnest; mantle thickest (about 84% volume); core densest and hottest.
Moho = crust/mantle · Gutenberg = mantle/outer core · Lehmann = outer/inner core.
Outer core liquid; inner core solid.

CHECK YOURSELF (answers at the end of the chapter)

Q1. Thickest layer: **mantle** · Q2. Waves that cannot pass liquids: **S-waves** · Q3. NIFE layer: **core**

CHAPTER 5 — PLATE TECTONICS, EARTHQUAKES AND VOLCANOES

Why does the ground shake and mountains rise?

5.1 The idea

IN SIMPLE WORDS (what / how / why)

The lithosphere is broken into big pieces called **tectonic plates** (7 major: Pacific, North American, South American, Eurasian, African, Indo-Australian (Indian + Australian), Antarctic). They move **a few centimetres a year**, about as fast as your fingernails grow.
Why do they move? Heat from the core makes the mantle rock flow very slowly in loops (**convection currents**): hot rock rises, cools, sinks. The plates ride on top.
Continental drift (Alfred Wegener, 1912): all continents were once joined as **Pangaea**; it split into **Laurasia (north) and Gondwanaland (south)**. **India was part of Gondwana** and moved north to hit Asia (about 50 million years ago) → **Himalayas**.

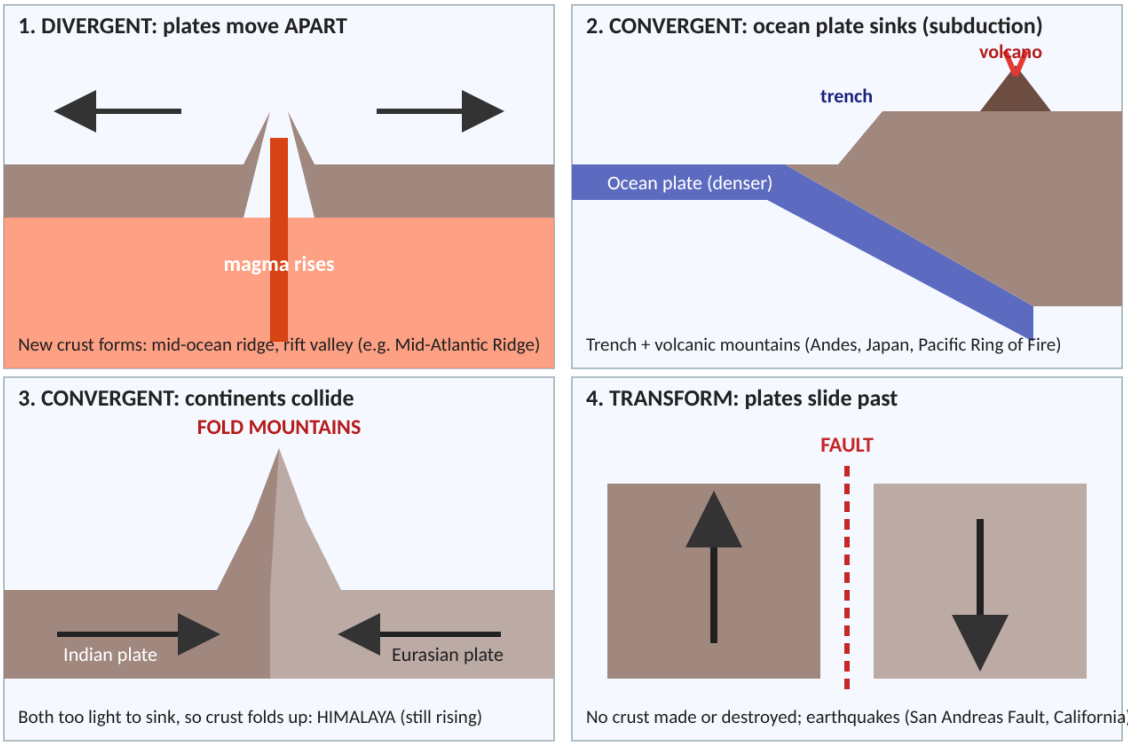


Fig 5.1 Four kinds of plate boundaries

5.2 Earthquakes

IN SIMPLE WORDS (what / how / why)

Why: plates are stuck together by friction while they keep trying to move. Stress builds up for years. Suddenly the rock breaks and slips (the **snap**), and the stored energy is released as **waves**.
Focus (hypocentre) = point underground where it starts. **Epicentre** = point on the surface directly above the focus; damage is highest here.
Measured by: **seismograph**; **Richter scale** (magnitude, energy; each +1 = about 10 times bigger shaking and about 32 times more energy), **Mercalli scale** (intensity, damage seen).

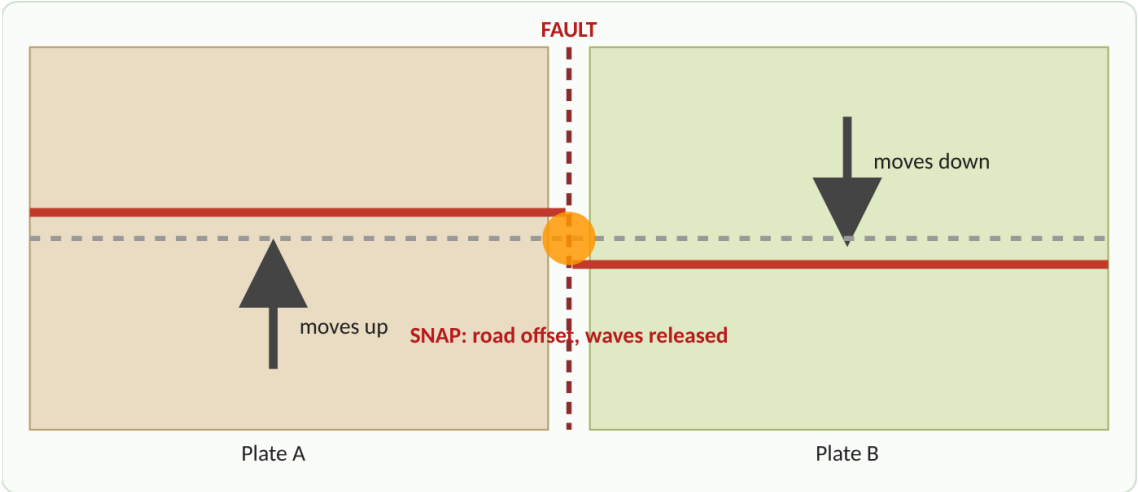


Fig 5.2 How an earthquake happens (stress builds, rock snaps, waves leave the focus)

Wave	Also called	Behaviour
P-wave	Primary / compressional	Fastest; pass through solid, liquid and gas; push-pull
S-wave	Secondary / shear	Slower; solids only; up-down shake
L-wave	Surface / long	Slowest; travel only on the surface; cause MOST damage

5.3 Volcanoes and the Ring of Fire

A **volcano** is an opening in the crust through which **magma** (underground) comes out as **lava** (on the surface), with gas and ash. Most are on plate boundaries.

Ring of Fire = horseshoe of volcanoes and earthquake zones round the **Pacific Ocean** (about 75% of the world's volcanoes, about 90% of earthquakes).
Types: active (erupted recently, e.g. Etna, Stromboli = 'Lighthouse of the Mediterranean', Barren Island), dormant, extinct.
India's only active volcano: Barren Island (Andaman). **Mauna Loa** (Hawaii) largest active; **Ojos del Salado** (Chile) highest active; **Mt Fuji** (Japan) dormant.

ANDHRA PRADESH / KURNOOL CONNECTION

Is Kurnool in an earthquake zone? India is split into zones II to V (BIS, 2002 map: Zone II, III, IV, V). Most of Andhra Pradesh lies in **Zones II and III** (low to moderate risk) because it sits on the old, stable **Deccan / Peninsular block**. Zone V is Kashmir, Himachal, Uttarakhand, North-East, Kutch and Andaman. (Detailed AP zoning: verify.)
Giant ancient rocks of AP: the **Eastern Ghats** and **Cuddapah basin** (Kurnool, Kadapa, Nandyal) are among the oldest rocks in India. This region is old and stable, which is why large earthquakes are rare.

EXAM TRAPS

Epicentre is on the surface; **focus** is underground.
Richter measures **magnitude**; Mercalli measures **intensity**.
India's only active volcano = Barren Island. **Deccan Traps** are an old lava (basalt) plateau, not an active volcano.
Tsunami (Japanese: harbour wave) is caused by undersea earthquake, not a storm.

2-MINUTE REVISION

Plates float on asthenosphere; move by convection.
Divergent = new crust; Convergent = mountain/trench; Transform = fault.
P fastest, S solids only, L most damage.
Ring of Fire = Pacific. Barren Island = India.

CHECK YOURSELF (answers at the end of the chapter)

Q1. Himalayas formed by: **continental–continental collision** · Q2. India's only active volcano: **Barren Island** · Q3. Surface wave: **L-wave**

CHAPTER 6 — ROCKS AND LANDFORMS

What is rock? How are mountains, plateaus and plains made?

6.1 Three kinds of rock

IN SIMPLE WORDS (what / how / why)

Igneous (fire-made): molten magma cools and hardens. Inside Earth (slow, big crystals) = **intrusive**, e.g. granite. On the surface (fast, small crystals) = **extrusive**, e.g. basalt.
Sedimentary (layer-made): rivers/wind carry broken rock (**sediments**) to lakes and seas; they settle in layers and get pressed into rock. Only these rocks contain **fossils**. Examples: sandstone, limestone, shale, coal.
Metamorphic (changed): any rock changed by **heat and pressure without melting**. Limestone → **marble**; shale → **slate**; granite → **gneiss**; sandstone → **quartzite**; coal → **graphite/diamond**.

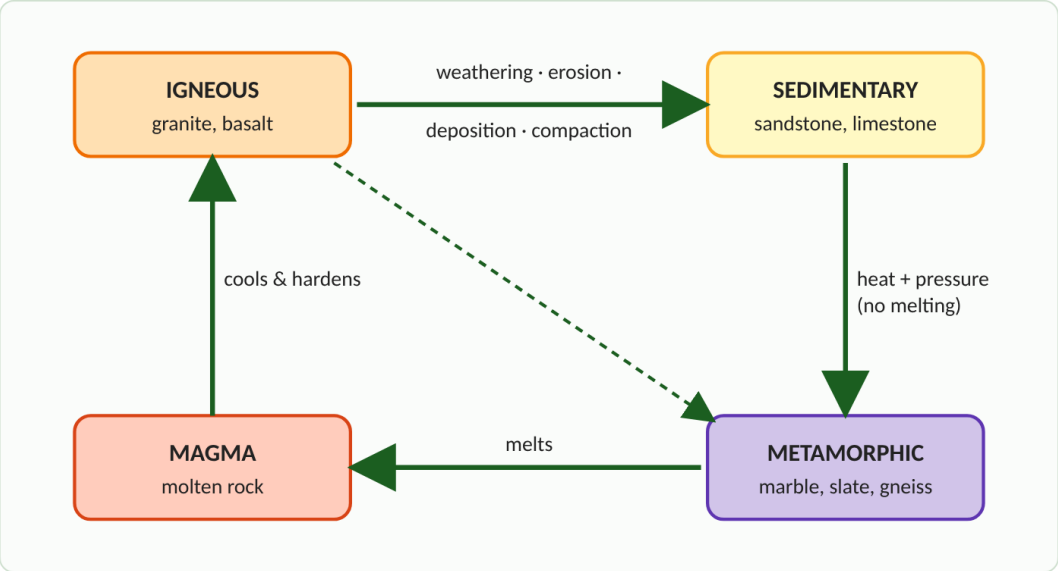


Fig 6.1 The rock cycle

ANDHRA PRADESH / KURNOOL CONNECTION

Kurnool is famous for rock! Banaganapalle and **Kurnool** limestone, **Kurnool group of rocks**, **Cuddapah system**, **Betamcherla** (Nandyal district) building stone/**Cuddapah slabs** (limestone) used for flooring, **Palnadu** limestone for cement. Cement factories in the Kurnool-Kadapa-Guntur belt use the limestone. Diamonds: historically **Wajrakarur (Anantapur)** and Banganapalle area (verify) — diamond-bearing kimberlite pipes of Anantapur-Kurnool.

6.2 Weathering and erosion

Weathering = rock breaks **in place** (by heat/cold, water, plants). **Erosion** = the broken material is **carried away** (by river, wind, glacier, sea). **Deposition** = it is dropped.
Rivers: upper course = V-valley, waterfall, gorge (erosion); middle = meanders; lower = **flood plain, levee, delta, oxbow lake** (deposition). **Delta** = triangular land where a river meets the sea (Godavari and Krishna deltas in AP).
Wind (deserts): sand dunes, **barchans** (crescent), mushroom rocks.
Glaciers: U-shaped valleys, moraines, **fiords** (sea inlets). **Sea waves**: cliffs, arches, stacks, beaches.

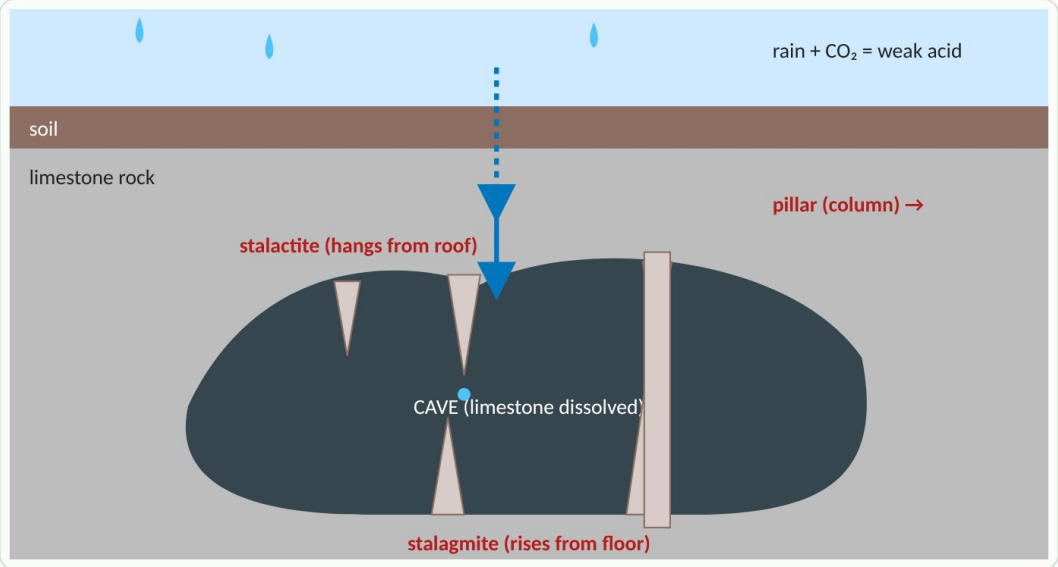


Fig 6.2 Limestone caves: how stalactites and stalagmites form

IN SIMPLE WORDS (what / how / why)

Cave story: rain water absorbs CO_2 and becomes weak acid → dissolves limestone → makes caves. Dripping water leaves tiny limestone each drip: hangs from roof = **stalactite**; grows up from floor = **stalagmite**; they join = **pillar**. Memory: **c**eiling = **c** (stalactite).

AP CONNECTION: CAVES

Borra Caves (Visakhapatnam, Ananthagiri hills): stalactite/stalagmite limestone caves. **Belum Caves** (Kurnool district, Kolimigundla): **second longest natural cave in the Indian subcontinent** (about 3.2 km; longest is Krem Liat Prah in Meghalaya, verify). **Yaganti** (Kurnool/Nandyal) cave temple, **Undavalli** caves (Guntur, rock-cut). These are favourite questions.

6.3 Landforms in short

Landform	What it is	Example
Mountain	High, steep, narrow top. **Fold** (Himalaya, Alps, Andes, Rockies), **Block/Fault** (Vosges, Black Forest), **Volcanic** (Fuji, Kilimanjaro), **Residual** (Aravalli, Eastern Ghats)	Aravalli = one of the oldest fold mountains
Plateau	High flat land with steep sides	Deccan, Tibet (highest), Chhotanagpur
Plain	Low flat land, usually made by river deposition	Indo-Gangetic, Coastal Andhra plains
Delta	Fan-shaped land at river mouth	Krishna-Godavari, Sundarbans

EXAM TRAPS

Eastern Ghats are **residual/dissected** hills, not young fold mountains.
Aravalli = **oldest fold mountain range** of India; Himalaya = **youngest**.
Fossils only in **sedimentary** rocks.

2-MINUTE REVISION

Igneous (granite, basalt) · Sedimentary (fossils, coal, limestone) · Metamorphic (marble, slate).

Landform	What it is	Example
Weather = in place; erosion = carried; deposition = dropped. Belum, Borra, Undavalli, Yaganti caves = AP.		

CHECK YOURSELF (answers at the end of the chapter)

Q1. Rock with fossils: **sedimentary** · Q2. Marble comes from: **limestone** · Q3. Belum caves are in: **Kurnool (Nandyal belt) district**

CHAPTER 7 — ATMOSPHERE

The air around us: layers, heat, pressure

IN SIMPLE WORDS (what / how / why)

Atmosphere = the blanket of gases held by Earth's gravity. About **78% nitrogen, 21% oxygen, 0.93% argon, 0.04% CO_2**, plus water vapour. Half of the air is within the lowest **5.5 km**, because gravity pulls air down: air gets **thinner as you go up** (that is why climbers need oxygen).

Why layers? Different heights absorb different amounts of the Sun's energy. Temperature goes down, then up, then down, then up again.

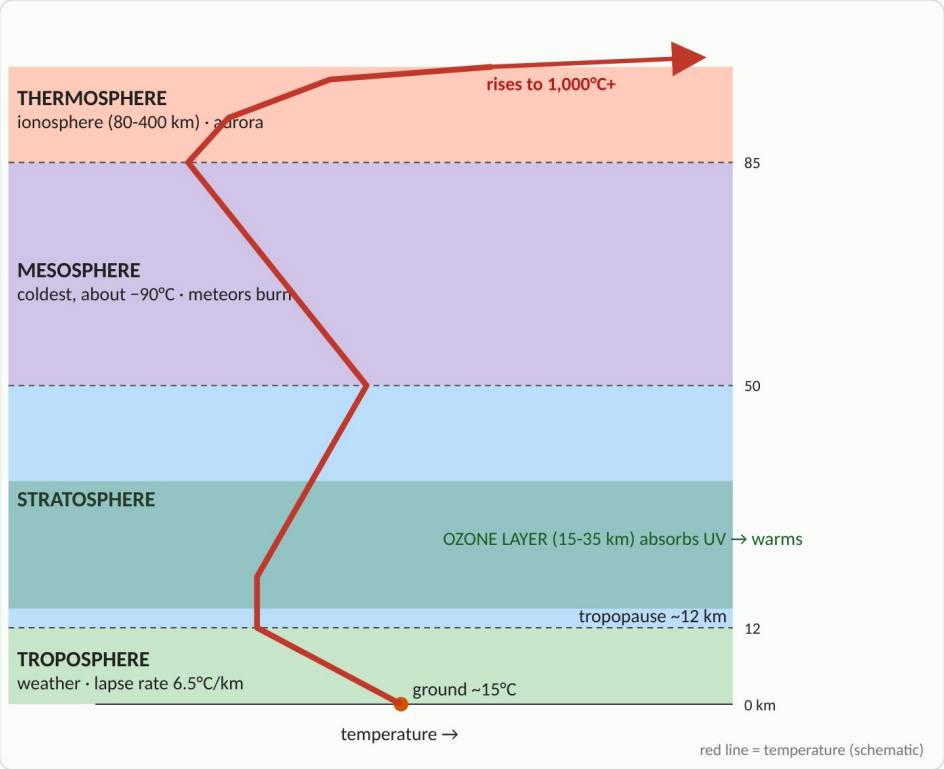


Fig 7.1 Layers of the atmosphere (red line = temperature)

Layer	Height	Remember
Troposphere	0-12 km (about 8 km at poles, 18 at equator)	Weather, clouds, rain, all life; temperature falls 6.5°C per km
Stratosphere	12-50 km	**Ozone layer** (15-35 km) absorbs UV; jet planes fly here; calm, no weather
Mesosphere	50-85 km	Coldest (about −90°C); meteors burn
Thermosphere	85-600 km	Very hot; **ionosphere** reflects radio waves; aurora; satellites
Exosphere	above	Fades into space

7.1 How is the air heated?

IN SIMPLE WORDS (what / how / why)

Sun's rays (**insolation**, short-wave) pass through air and heat the **ground** first. The ground then sends out **long-wave heat (terrestrial radiation)**, which **heats the air from below**. So the air is warmer near the ground.

Greenhouse effect: gases like CO_2, methane, water vapour trap this outgoing heat, like glass in a greenhouse. A natural greenhouse keeps Earth at about **15°C** instead of −18°C. The problem is the **extra** gases from burning fuels.

Heat budget: of 100 units of sunlight, about **35 are reflected back** (albedo: clouds, snow), 14 are absorbed by the atmosphere and 51 reach the ground.

Ozone hole: CFCs destroy ozone → **Montreal Protocol (1987)**; Kigali Amendment (2016) limits HFCs. **Kyoto Protocol (1997)** and **Paris Agreement (2015)** deal with global warming (not ozone).
Pressure: weight of the air above; measured by **barometer**; unit millibar (mb) or hectopascal. Normal at sea level about **1013 mb**. It falls as you go up.

ANDHRA PRADESH / KURNOOL CONNECTION

AP gets very hot in **April-May** (Kurnool/Nandyal and the coastal belt around Vijayawada often cross 42-45°C; verify exact records). Heat waves and **lightning deaths** are major hazards (APSDMA warnings, 'Lightning alert' apps).

EXAM TRAPS

Ozone is in the stratosphere; weather is in the troposphere.
Montreal = ozone; Kyoto/Paris = climate.
Air is heated by the **ground**, not directly by the sun, so mountain tops are colder.

2-MINUTE REVISION

N 78 · O 21 · Ar 0.93 · CO_2 0.04.
Troposphere weather · Stratosphere ozone · Mesosphere coldest · Thermosphere ionosphere.
Lapse rate 6.5°C/km.

CHECK YOURSELF (answers at the end of the chapter)

Q1. Layer with ozone: **stratosphere** · Q2. Coldest layer: **mesosphere** · Q3. Normal sea-level pressure: **1013 mb**

CHAPTER 8 — WINDS, MONSOON AND RAIN

Why does air move? Why does it rain?

IN SIMPLE WORDS (what / how / why)

Wind = air moving from high pressure to low pressure. Why? Warm air is light, rises and leaves **low pressure**. Cool air is heavy, sinks and creates **high pressure**. Air always flows to fill low pressure, like water flowing downhill. The bigger the pressure difference, the stronger the wind.

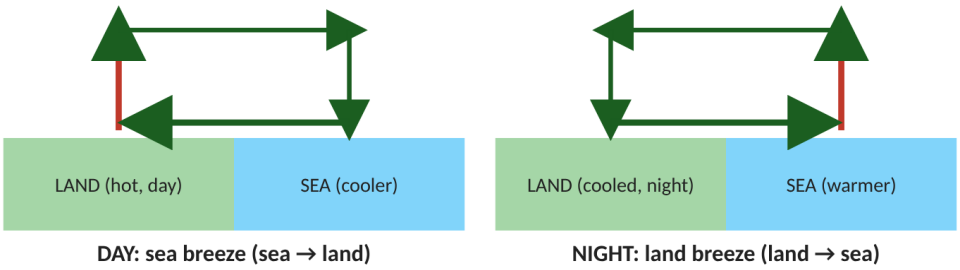


Fig 8.1 Sea breeze (day) and land breeze (night): the same idea on a small scale

8.1 Big winds of the world

IN SIMPLE WORDS (what / how / why)

Because Earth rotates, winds do not go straight: they bend **right in the northern hemisphere and left in the southern hemisphere** (**Coriolis effect**).
Trade winds (0°-30°): steady, blow towards the equator from the east (**NE trades in the north, SE in the south**). **Westerlies** (30°-60°): from the west; strong in the south = 'Roaring Forties, Furious Fifties, Screaming Sixties'. **Polar easterlies** (60°-90°).

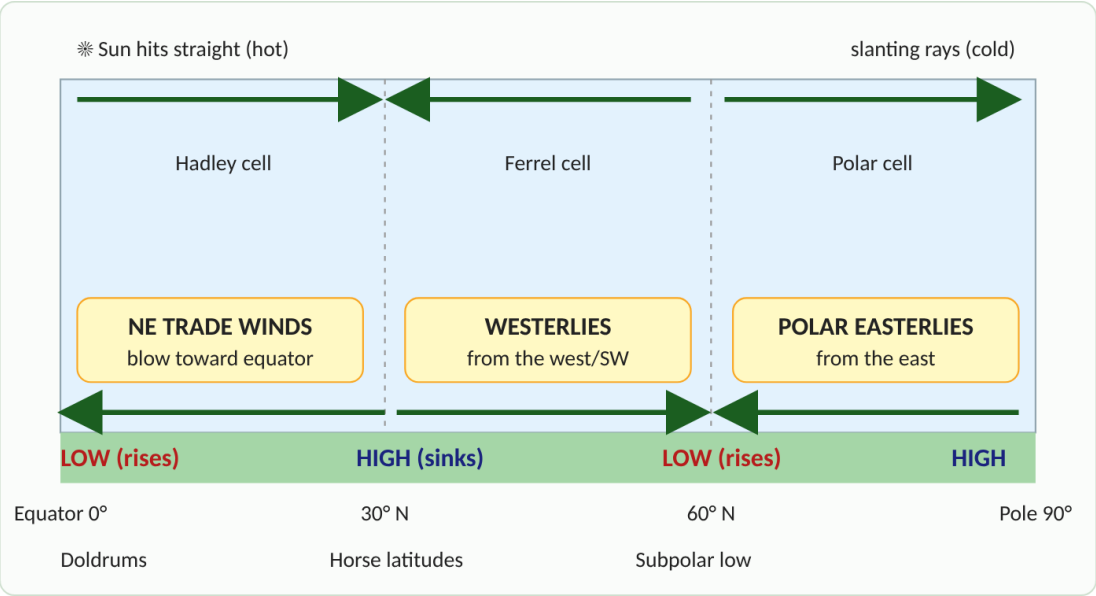


Fig 8.2 Global wind belts and the Coriolis bend

8.2 Local winds (frequent in exams)

Wind	Place	Type
Loo	North India/Pakistan plains (summer)	Hot, dry, dusty; causes heat waves
Chinook	Rockies (USA/Canada)	Warm, dry; 'snow eater'
Foehn	Alps	Warm, dry
Mistral	France	Cold, from Alps
Harmattan	West Africa	Hot, dry, dusty; 'Doctor wind'
Sirocco	Sahara to Mediterranean	Hot, dusty
Kal Baisakhi	West Bengal, Assam (Apr-May)	Thunderstorm; Norwesters
Mango showers	Kerala, Karnataka	Pre-monsoon rain

8.3 Rain: three types

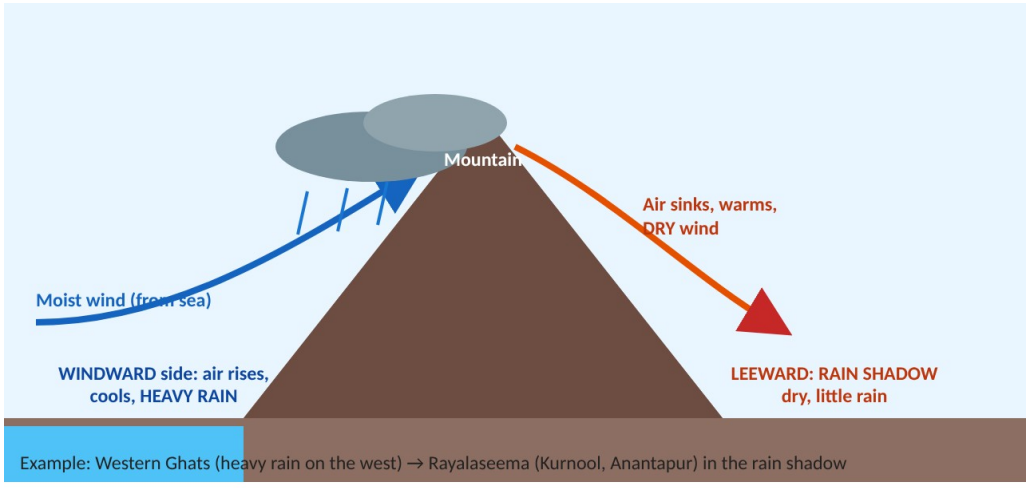


Fig 8.3 Relief (orographic) rain and rain shadow

Convectional: ground heats → air rises → cools → clouds → rain (daily in equatorial regions).
Orographic (relief): wind is forced up a mountain → rain on the windward side; dry rain shadow on the leeward side.
Cyclonic (frontal): when warm and cold air masses meet (see Chapter 9).

8.4 Indian monsoon (basics)

IN SIMPLE WORDS (what / how / why)

Monsoon (Arabic *mausim* = season): winds that **reverse direction** with the season. **Why:** in summer land heats faster than the sea → low pressure over north-west India → moist **south-west** winds rush in from the Indian Ocean = rain (June-Sept). In winter the land is cold → high pressure → dry **north-east** winds flow out to the sea.

Two branches: **Arabian Sea branch** (hits Western Ghats first, heavy rain on the west coast: Kerala onset about 1 June) and **Bay of Bengal branch** (Northeast India, Gangetic plain). **Retreating monsoon** (Oct-Nov) = 'October heat', and **Tamil Nadu coast and south coastal AP get most rain from the North-East monsoon (Oct-Dec)**.
El Niño (warm Pacific off Peru) weakens the Indian monsoon → **drought**; **La Niña** brings good rain. **IOD (Indian Ocean Dipole)** positive helps the monsoon.



Fig 8.4 El Niño

ANDHRA PRADESH / KURNOOL CONNECTION

Rayalaseema (Kurnool, Anantapur, Kadapa, Chittoor) lies in the **rain shadow of the Western Ghats**: only about **500-700 mm** rain a year (Anantapur is among the driest districts of peninsular India). The coastal districts get more from the NE monsoon and cyclones. **That is why droughts, tank irrigation and the Handri-Neeva / Galeru-Nagari projects matter for Kurnool.**
AP receives rain from both: SW monsoon (about 60% of rain; Jun-Sep) and NE monsoon (Oct-Dec; significant for coastal and southern AP).

EXAM TRAPS

Monsoon rain on the leeward side is low, not high.
Wind direction is named by where it **comes from** (south-west monsoon = from SW).
El Niño = drought risk for India; La Niña = better rain.

2-MINUTE REVISION

Wind: high → low pressure. Coriolis: right (north), left (south).
Trade winds, Westerlies, Polar easterlies.
Rain: convectional, orographic, cyclonic. Rayalaseema = rain shadow.

CHECK YOURSELF (answers at the end of the chapter)

Q1. Hot wind of north Indian plains: **Loo** · Q2. Region of AP in rain shadow: **Rayalaseema** · Q3. El Niño causes: **weak monsoon**

CHAPTER 9 — FRONTS, CYCLONES AND CLIMATE

Weather vs climate; storms; types of climate

9.1 Air masses and fronts

IN SIMPLE WORDS (what / how / why)

Front = the boundary where a **warm** air mass meets a **cold** one. They do not mix easily. **Warm air is lighter, so it is pushed up** over the cold air → cooling → clouds → rain.
Cold front: cold air pushes under warm air; short, heavy, sudden rain, thunder. **Warm front**: warm air slides over cold; long, light rain.

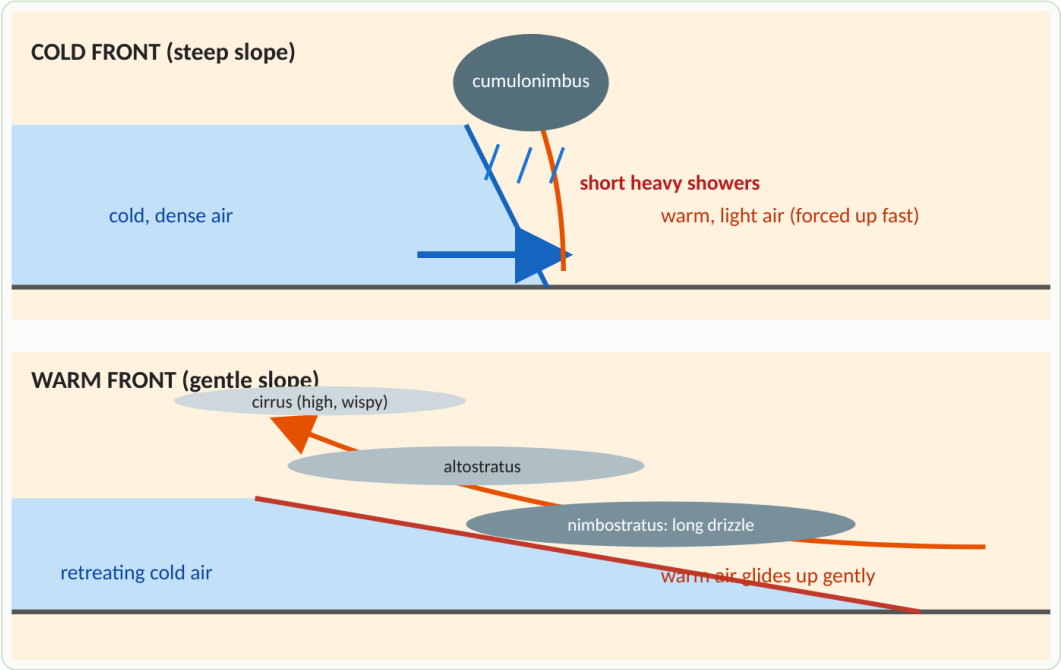


Fig 9.1 Fronts and Western Disturbances

ANDHRA PRADESH / KURNOOL CONNECTION

Western Disturbances are winter storms from the Mediterranean that give rain/snow to north India (good for the **rabi crops**, like wheat). They do not reach AP much.

9.2 Tropical cyclones

IN SIMPLE WORDS (what / how / why)

What: a huge spinning storm over **warm tropical seas**. **How:** sea above **26.5°C** → water evaporates (fuel) → warm moist air rises fast → low pressure at the surface → air rushes in and, due to Coriolis, **spins (anticlockwise in the northern hemisphere, clockwise in the southern)**. The rising vapour condenses and gives out heat, making the storm stronger. The calm centre = **eye**; the strongest winds = **eye wall**.
Why do cyclones die on land? No warm water (no fuel), and friction slows them.
Names: Hurricane (Atlantic, Caribbean), **Cyclone (Indian Ocean)**, Typhoon (China Sea), Willy-willy (Australia). **Storm surge** (sea rushing onto coast) kills most people.

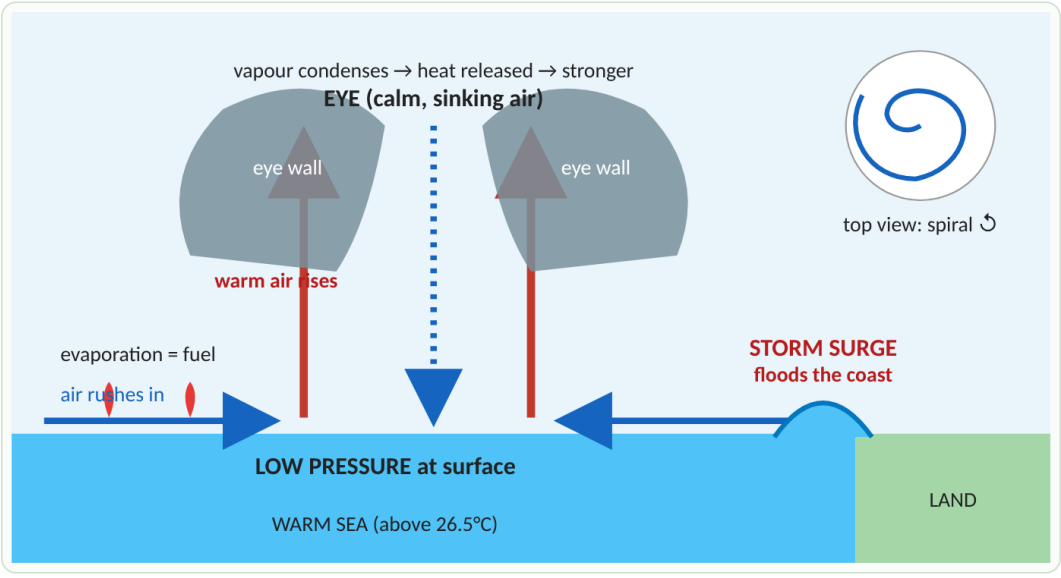


Fig 9.2 Cross section of a tropical cyclone

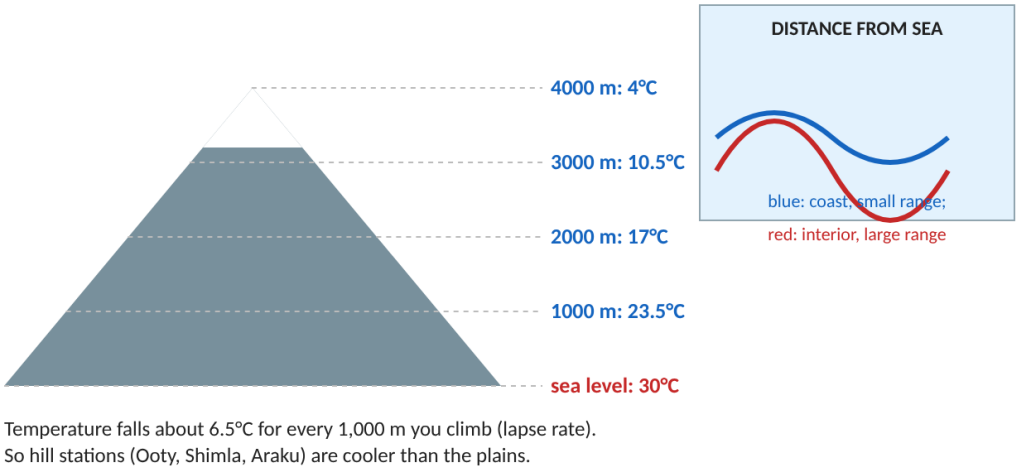
AP CONNECTION: CYCLONES

AP coast is hit most often in **Oct-Nov** (post-monsoon) from the **Bay of Bengal** (more cyclones than the Arabian Sea). Famous: Hudhud (2014, Visakhapatnam), Titli (2018, Srikakulam), Phailin (2013), Michaung (Dec 2023, Bapatla/Nellore), Montha (Oct 2025). Early-warning agency: **IMD**; state: **APSDMA**. Names given in rotation by the WMO/ESCAP list agreed by 13 countries

(India, Bangladesh, Maldives, Myanmar, Oman, Pakistan, Qatar, Saudi Arabia, Sri Lanka, Thailand, UAE, Yemen, Iran).

9.3 Weather and climate

Weather = atmosphere's condition **today/this week** at a place. **Climate** = **average of weather over about 30 years**.
Factors that decide climate: (1) **Latitude** (2) **Altitude** (3) **Distance from sea** (4) **Ocean currents** (5) **Winds** (6) **Relief (mountain direction)** (7) **Forest/soil**.



Temperature falls about 6.5°C for every 1,000 m you climb (lapse rate).
So hill stations (Ooty, Shimla, Araku) are cooler than the plains.

Fig 9.3 Altitude and distance from sea

Type (Köppen idea)	Where	Key point
Tropical rainforest (Af)	Equator: Amazon, Congo	Hot, rains daily, evergreen forest
Savanna (Aw)	Africa, Brazil highlands	Wet and dry seasons; grassland; **India's monsoon type is Am/Aw**
Hot desert (BW)	Sahara, Thar	Very little rain; large day-night difference
Mediterranean (Cs)	California, Chile, Mediterranean	Dry summer, wet winter; citrus, grapes
Humid subtropical (Cfa)	SE China, SE USA	Hot humid summer
Tundra / Polar	Siberia, Greenland, Antarctica	Very cold, ice-cap

ANDHRA PRADESH / KURNOOL CONNECTION

Kurnool's climate is **tropical semi-arid (steppe)**: hot summers, moderate winters, low and uncertain rain. **Rayalaseema = semi-arid**; **coastal Andhra = tropical wet-and-dry**.

EXAM TRAPS

Cyclone spin: anticlockwise in north, clockwise in south.
Weather ≠ climate (time period).
Bay of Bengal cyclones hit AP/Odisha/WB/TN; **Arabian Sea** ones hit Gujarat/Maharashtra/Kerala.

2-MINUTE REVISION

Front = warm meets cold. Cyclone needs sea above 26.5°C and Coriolis.
Weather = short term; Climate = 30 years.
Rayalaseema semi-arid; coast gets Oct-Nov cyclones.

CHECK YOURSELF (answers at the end of the chapter)

Q1. Minimum sea temperature for cyclones: **26.5°C** · Q2. Climate = average of: **about 30 years** · Q3. Cyclone rotation in north: **anticlockwise**

CHAPTER 10 — OCEANS: CURRENTS, TIDES, RESOURCES

Why does the sea move?

10.1 Oceans at a glance

Ocean	Remember
Pacific	Largest and deepest; **Mariana Trench (Challenger Deep) about 11,000 m** deepest point on Earth; Ring of Fire
Atlantic	S-shaped; busiest trade; Sargasso Sea (no coast); Puerto Rico Trench

Ocean	Remember
Indian	Only ocean named after a country; its northern part is closed by Asia → **monsoon reversal of currents**; Java (Sunda) Trench deepest here
Southern	Around Antarctica
Arctic	Smallest and shallowest

10.2 Ocean currents

IN SIMPLE WORDS (what / how / why)

What: big rivers inside the sea. **Why they move:** (1) **prevailing winds** push the surface water; (2) **Earth's rotation** (Coriolis) bends them; (3) **temperature and salt differences**; (4) **land blocks** them, so they curve and form **gyres** (circular loops).
Result: in the northern oceans gyres turn **clockwise**, in the southern **anticlockwise**. **Warm currents** flow from the equator to the poles and make the coast **warmer and wetter** (Gulf Stream warms Europe). **Cold currents** flow from the poles and make coasts **cooler and drier** (Peru and Benguela currents → Atacama and Namib deserts).
Where warm and cold meet = **fogs** and rich **fishing grounds** (Grand Banks, Newfoundland; North Sea; north-east Japan).

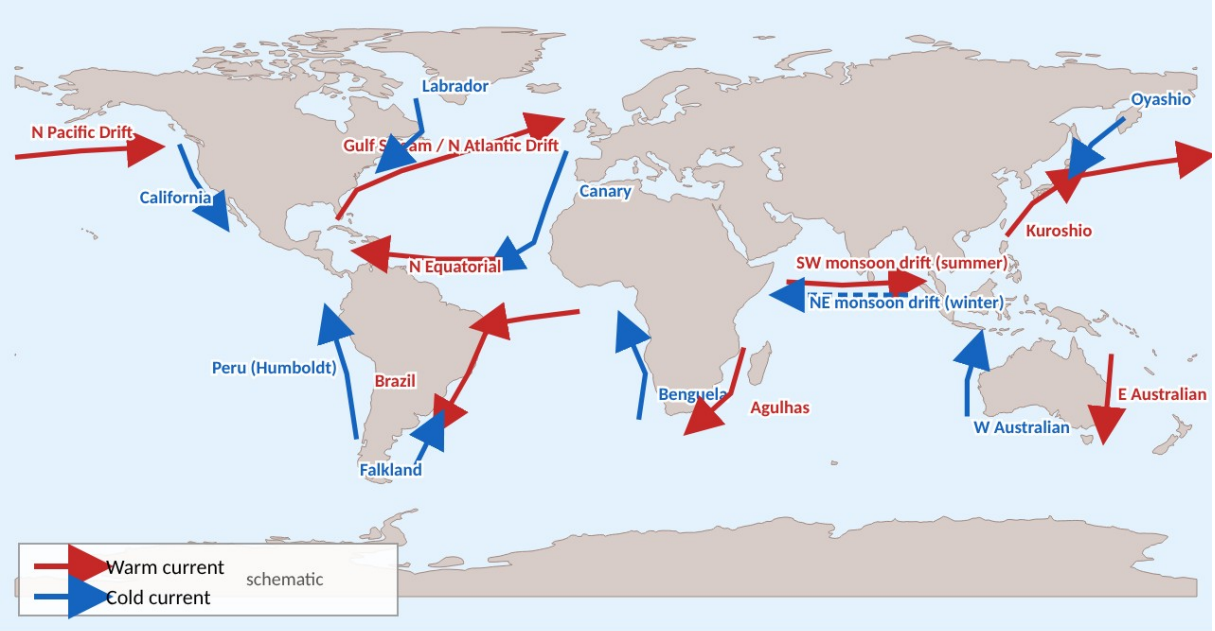


Fig 10.1 Major currents (schematic): red = warm, blue = cold

ANDHRA PRADESH / KURNOOL CONNECTION

Indian Ocean is special: currents **reverse with the monsoon** (the only ocean where this happens). **Coromandel coast** (AP + Tamil Nadu) gets rain from the **North-East monsoon** when the winds blow over the Bay of Bengal. Fishing coast of AP (Kakinada, Machilipatnam, Visakhapatnam, Nellore): **2nd-3rd longest coastline of Indian states (about 1,053 km; verify)**.

10.3 Tides

IN SIMPLE WORDS (what / how / why)

Tide = rise and fall of the sea level twice a day, caused by the **pull of the Moon (main) and the Sun**. The Moon's pull makes a bulge on the side facing it, and another bulge opposite (because of Earth's spin).
When Sun, Moon and Earth are in a **straight line** (new/full moon) the pulls add → **spring tide (highest)**. At **right angles** (quarter moons) the pulls partly cancel → **neap tide (lowest)**.

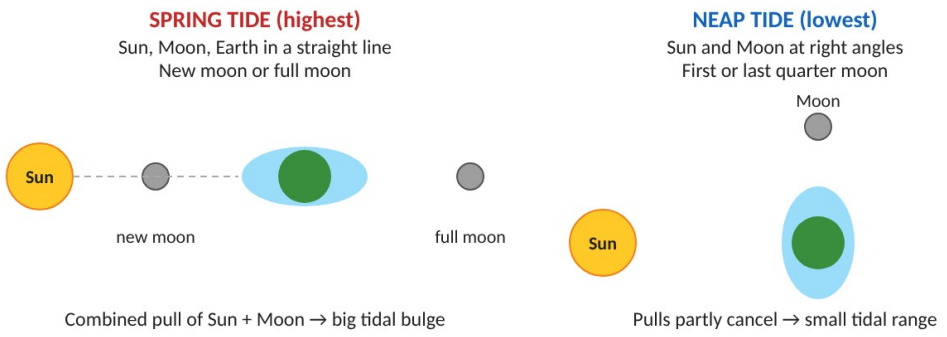


Fig 10.2 Spring and neap tides

Uses: navigation (ports like **Kolkata** use tides), **tidal power**, fishing. **India's tidal power potential**: Gulf of Khambhat, Gulf of Kutch, Sundarbans.
Salinity: average about **35 parts per thousand (35‰)**; highest in the **Dead Sea/Great Salt Lake**; Red Sea very high; lowest near river mouths and the Baltic Sea.
Coral reefs need warm, clear, shallow water (**above 20°C**): Great Barrier Reef (Australia), India: **Gulf of Mannar, Gulf of Kutch, Lakshadweep, Andaman-Nicobar**. **Bleaching** = corals lose algae due to warm water.

AP CONNECTION: COAST

Pulicat Lake (Tirupati/Nellore–Tamil Nadu border) = **India's second largest brackish-water lagoon after Chilika** (Odisha); bird sanctuary. **Kolleru Lake** (Eluru/Krishna) = **largest freshwater lake in AP, Ramsar site**. **Krishna-Godavari basin** = major offshore gas. **Ports**: Visakhapatnam (major, national), Krishnapatnam, Kakinada (deep-water), Gangavaram, Ramayapatnam; Machilipatnam (new, planned/under development; verify).

EXAM TRAPS

Tides are caused by the **Moon mainly**, not by wind.
Spring tide has **nothing to do with the spring season**.
Chilika = largest brackish lagoon in India (and Asia); **Pulicat = second**.
Deepest point: Mariana Trench (Pacific), not the Atlantic's Puerto Rico Trench.

2-MINUTE REVISION

Currents: wind + rotation + temperature + land; N gyres clockwise.
Warm = wetter coast; cold = desert coast.
Spring tide = new/full moon; Neap = quarter moon.
Pulicat (2nd) vs Chilika (1st).

CHECK YOURSELF (answers at the end of the chapter)

Q1. Highest tides occur on: **new moon and full moon** · Q2. Cold current of Peru: **Humboldt** · Q3. Largest lagoon in India: **Chilika**

CHAPTER ★ — FINAL SELF-TEST: 30 MIXED QUESTIONS

Take after finishing all 10 chapters. Cover the answers (given after Q30).

#	Question	Answer
1	Which planet has the highest surface temperature?	**Venus**
2	The seasons are caused by:	**Tilt of Earth's axis along with revolution**
3	Earth is nearest the Sun on about:	**3 January (perihelion)**
4	Tropic of Cancer passes through which state?	**Madhya Pradesh (not AP)**
5	IST is calculated from:	**82°30′E**
6	Kurnool (78°E) local time compared to IST:	**About 18 minutes behind**
7	Thickest layer of the Earth:	**Mantle**
8	S-waves cannot travel through:	**Liquid (outer core)**
9	Himalayas were formed by collision of:	**Indian plate and Eurasian plate**
10	India's only active volcano:	**Barren Island**
11	Which rock contains fossils?	**Sedimentary**
12	Marble is made from:	**Limestone (metamorphic)**

#	Question	Answer
13	Belum Caves are in:	**Kurnool/Nandyal region of AP**
14	Ozone layer lies in:	**Stratosphere**
15	Coldest atmospheric layer:	**Mesosphere**
16	Montreal Protocol is about:	**Ozone-depleting substances (CFCs)**
17	Winds bend right in the northern hemisphere due to:	**Coriolis effect**
18	Rayalaseema gets low rain because it lies in:	**Rain shadow of the Western Ghats**
19	El Niño usually brings India:	**Weak monsoon, drought risk**
20	Minimum sea temperature for a cyclone:	**26.5°C**
21	Cyclones in the Northern Hemisphere rotate:	**Anticlockwise**
22	Season when AP coast gets most cyclones:	**Oct-Nov (post monsoon)**
23	Climate is average weather of about:	**30 years**
24	Warm currents make coastal climate:	**Warmer and wetter**
25	Desert formed beside Peru current:	**Atacama**
26	Highest tide:	**Spring tide (new/full moon)**
27	Largest brackish lagoon of India:	**Chilika (Pulicat is second)**
28	Deepest point on Earth:	**Challenger Deep, Mariana Trench**
29	Largest freshwater lake of AP (Ramsar):	**Kolleru**
30	ISRO's launch site in AP:	**Sriharikota (Satish Dhawan Space Centre)**

SCORE AND NEXT STEP

How did you score? 24+ = strong. 18-23 = good, re-read the green boxes. Below 18 = re-read the blue boxes of the chapters you missed. Mark the questions you got wrong; send them to me and I will make a targeted revision set.
Next volume: Geography Vol 2 — Physical Geography of India (Himalayas, plains, plateau, rivers, soils, vegetation).